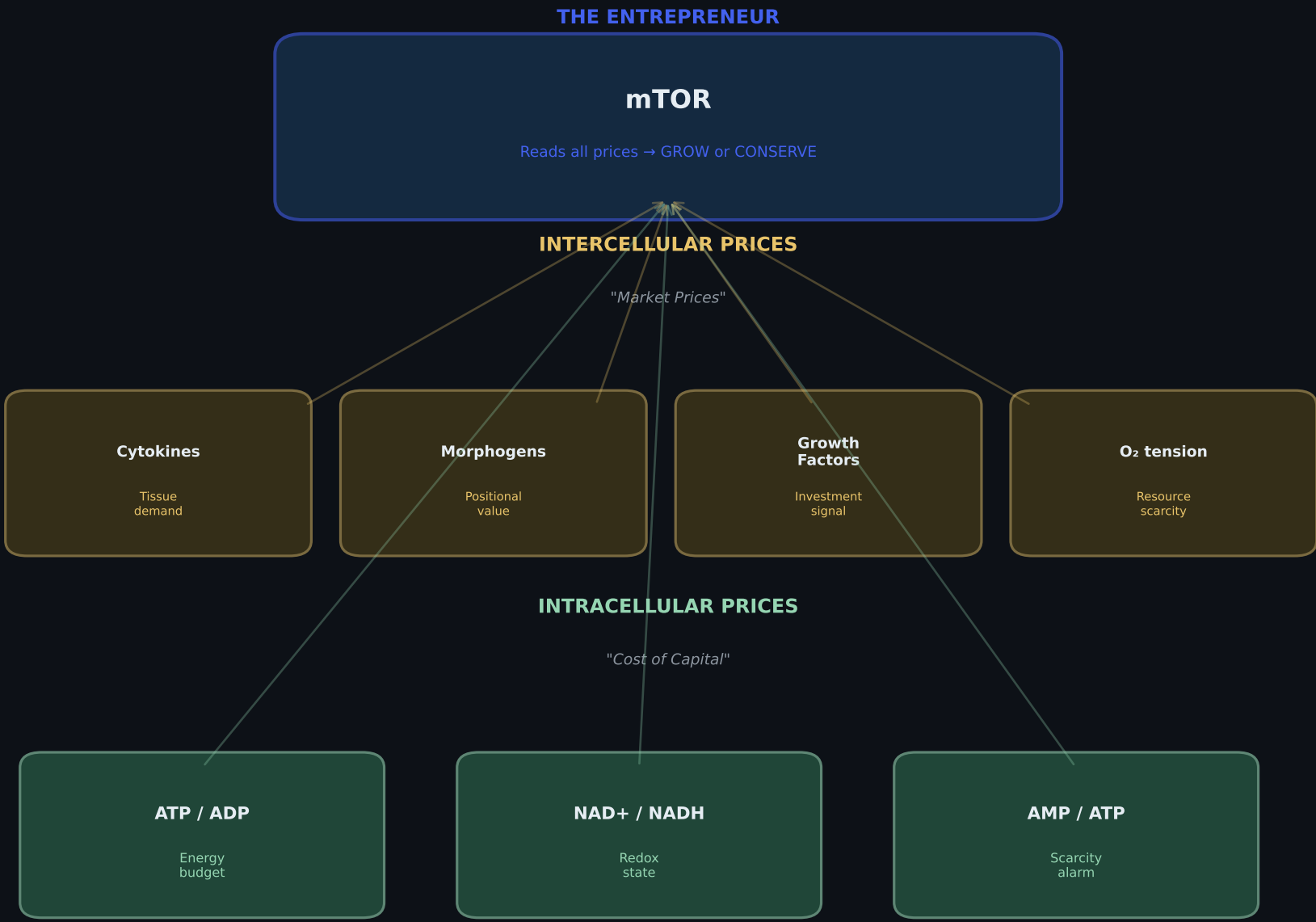


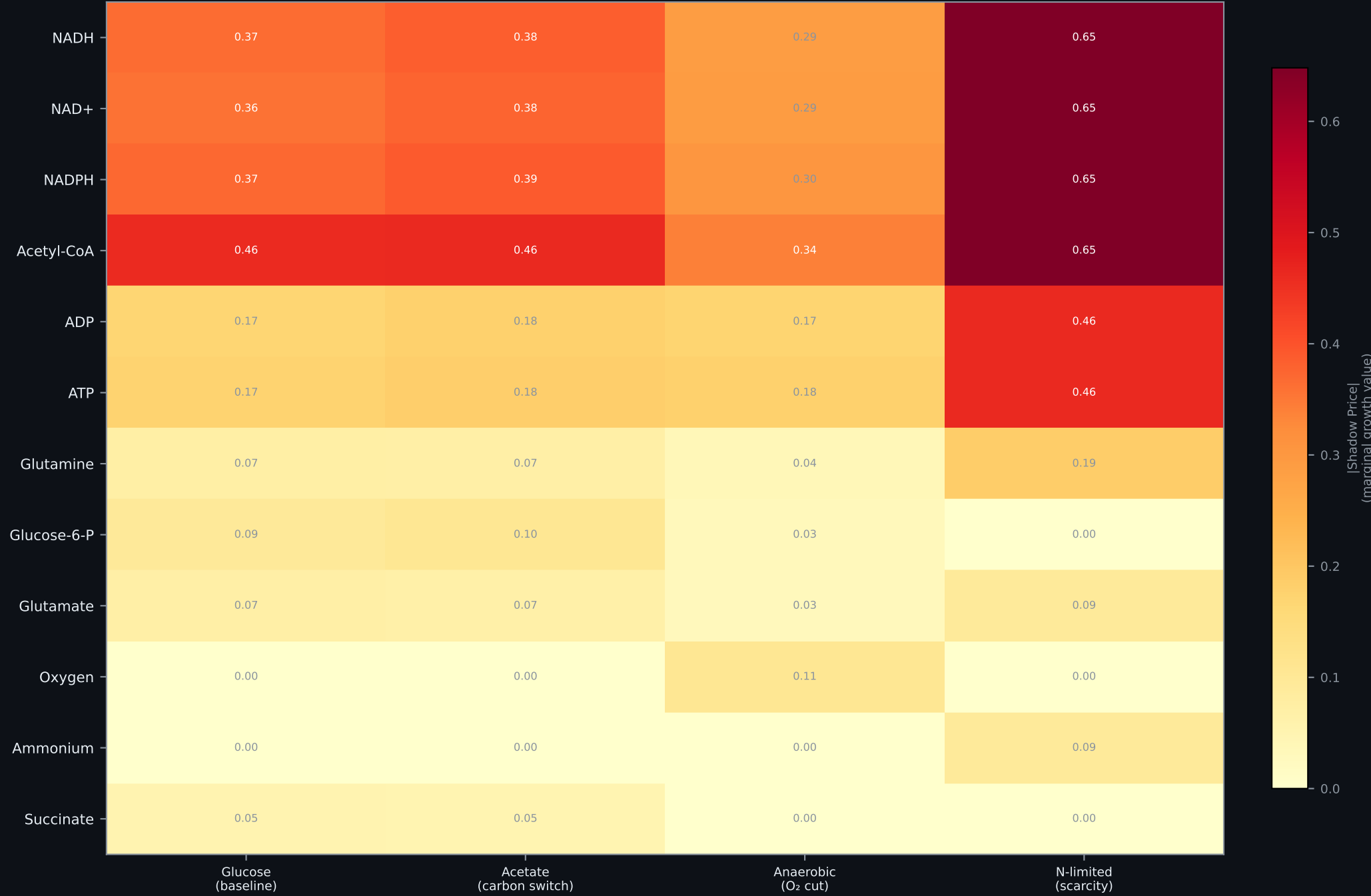
A

The Price System of the Cell: Life as a Decentralized Economy

The Price System of the Cell



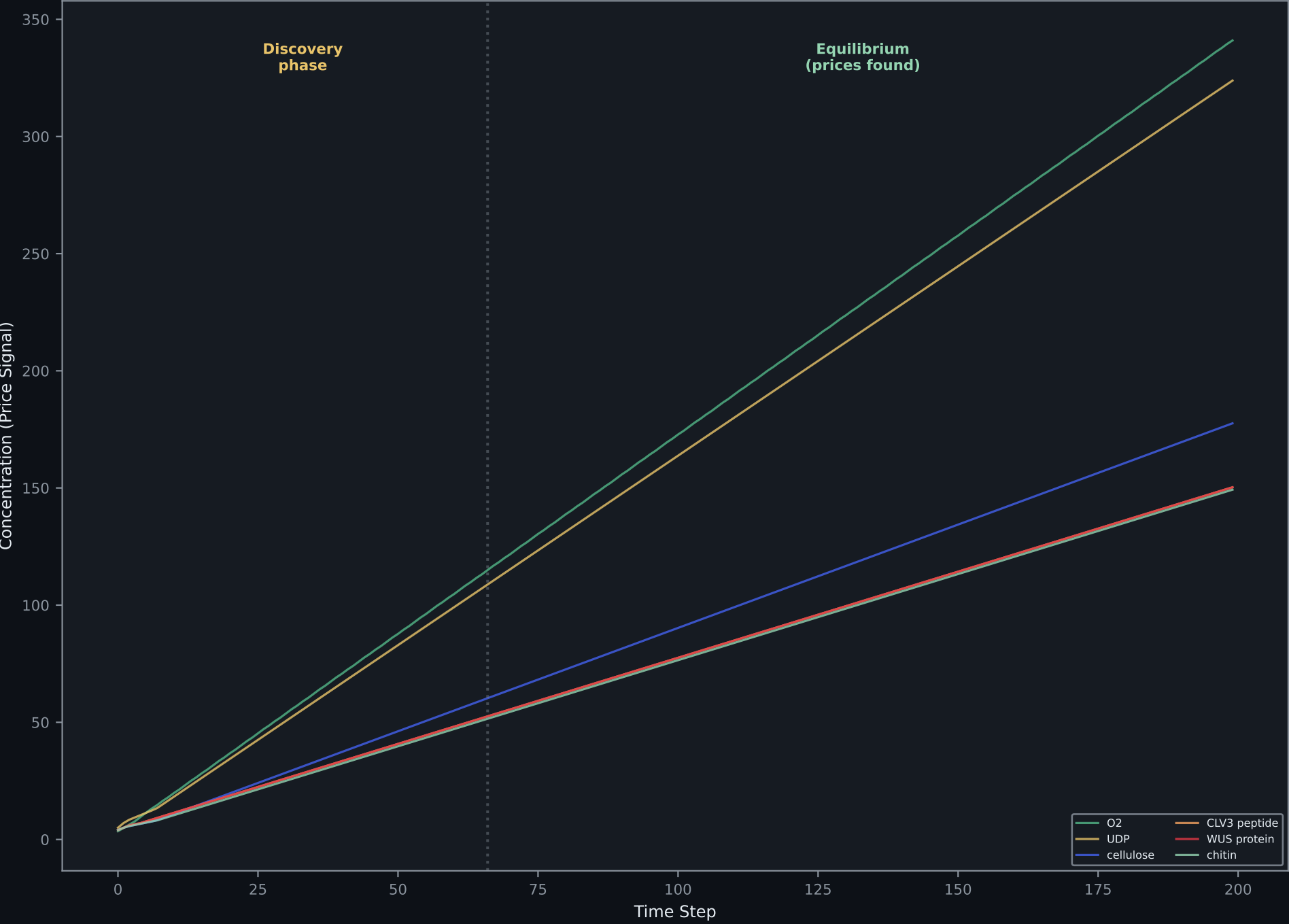
Shadow Prices Shift With Context
(same metabolite, different value)



Menger's subjective value: the price of a metabolite depends on the environment the cell is in, not on the molecule itself.

C

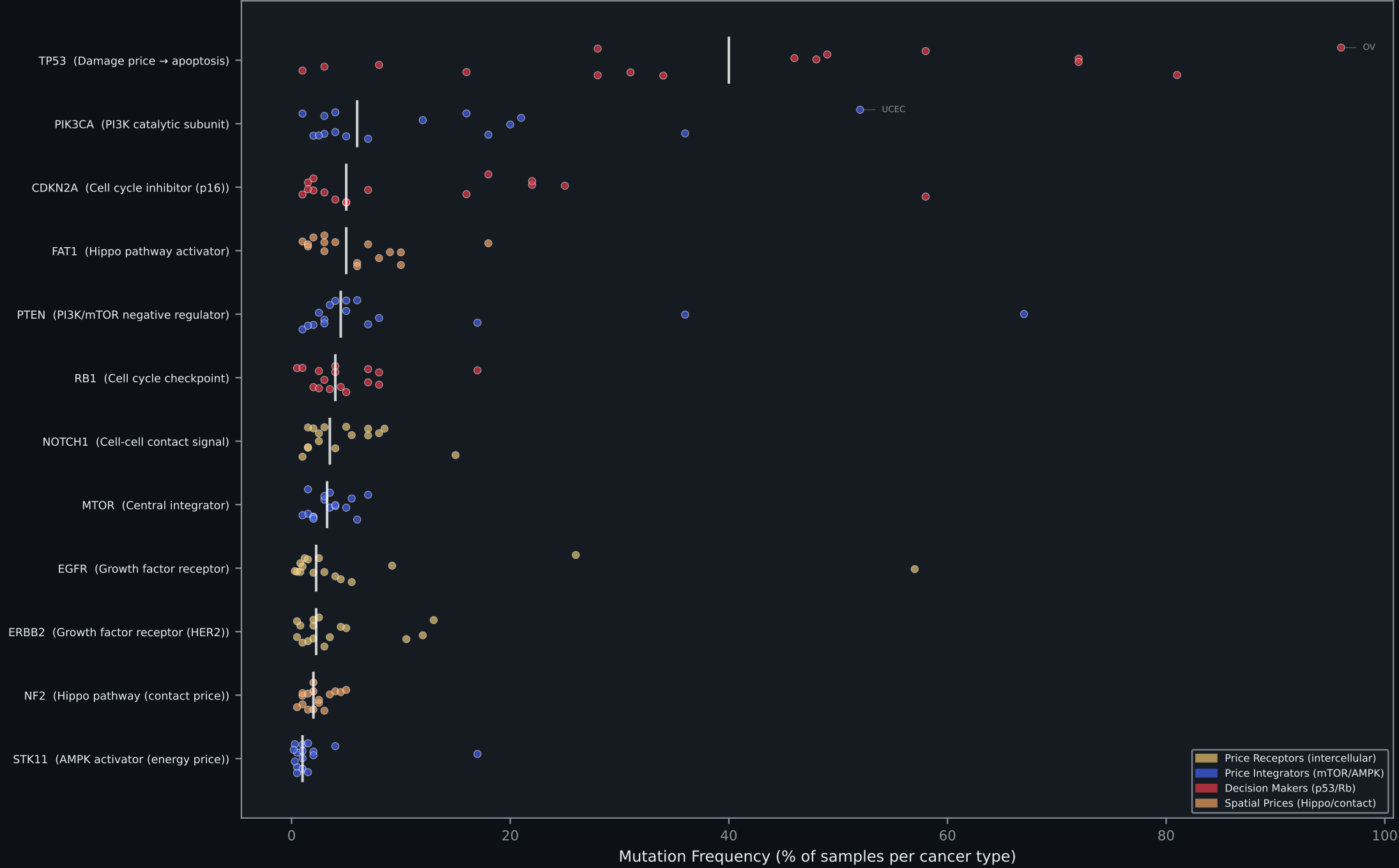
Price Discovery in the Distributed Economy
(metabolite pools oscillate then converge)



Oversupply → producers slow (price drops). Scarcity → producers speed up (price rises).
No planner needed. The invisible hand finds equilibrium.

D

Cancer Mutations Target the Price System
(every dot = one cancer type, white line = median)



The variation IS the signal: TP53 is 96% in ovarian but 1% in thyroid.
Different tissues break different price components. Cancer is tissue-specific misallocation.